

Yuvraj Pratiksh Raghuvanshi

Final Year Postgraduate, Department of Computer Science, Indian Institute of Technology, Kanpur

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ACADEMIC QUALIFICATIONS

Degree/Certificate	Institute	CPI/%	Year
M.Tech.	Indian Institute of Technology, Kanpur	8.57/10	2024 – Present
B.Tech.	ABES Institute of Technology, Ghaziabad	7.59/10	2018 – 2022
CBSE(XII)	J. P. Public School, Baraut (Private Candidate)	72.6%	2018
CBSE(X)	J. P. Public School, Baraut	8.0/10	2015

WORK EXPERIENCE

- **Founding Engineer** — Nyay Setu AI Pvt. Ltd., Kanpur (Jan 25 – Sep 25)
 - Early contributor to a **legal-tech platform** connecting lawyers and clients via chat, calls, and appointments.
 - Architected **infrastructure** with optimized schemas, scalable APIs, and modular UI.
 - Built **real-time systems** using Firebase (messaging) and Agora SDK (video/voice).
 - Prototyped **AI tools** for legal summarization and case guidance; now a silent partner.**Keywords:** *Firebase, Agora SDK, WebRTC, PostgreSQL, Project Management, Web Application, AI for Law*
- **Software Engineer** — Sixdis Tech. Pvt. Ltd. (Oolka.in), Bangalore (Dec 2023 – May 2024)
 - Worked on the credit-analysis platform *Oolka*, after Sixdis acquired its IP from Rupymart.
 - Ensured smooth **product transition** by maintaining API versions, adding features, and supporting deployments.**Keywords:** *API Development, Python, Django, FinTech, Product Transition, System Design*
- **Software Engineer** — Rupymart Tech. Pvt. Ltd., Kolkata (Apr 2023 – Nov 2023)
 - Built infrastructure: RESTful APIs, PostgreSQL schemas, CI/CD pipelines, AWS + Docker deployments.
 - Automated integration of credit bureau data to generate and analyze credit reports.
 - Extended the bank statement parser  Inkredo with Django, pdfplumber, regex, and ML models.
 - **Migrated entire third-party JavaScript codebase to Python3** for scalability and in-house ownership.**Keywords:** *Python, FastAPI, PostgreSQL, AWS, Docker, ML Integration, System Migration*
- **Software Engineer** — Vectoscalar Tech. Pvt. Ltd., Noida (Nov 2021 – Oct 2022)
 - Built scalable microservices using Python Flask for e-commerce, **ensuring 99.9% uptime** on AWS.
 - Cut infrastructure costs with **AWS Lambda** and optimized deployments via **Docker + CI/CD pipelines**.
 - **Designed MongoDB schemas** for reliable, high-volume transactions with seamless scalability.**Keywords:** *Python Flask, Microservices, AWS Lambda, CI/CD, Docker, MongoDB, Cloud Infrastructure*

RESEARCH WORK

- **Poetry Generation in Hindi** — Anviksha Labs, BharatGen  – Prof. Arnab Bhattacharya, IIT Kanpur (May 2025 – Present)
 - Curating & preprocessed large-scale datasets of Hindi poems, quotes, dohas, and folk literature for training.
 - Fine-tuned **IndicBERTv2** and related models for multi-class poet identification and multi-label thematic classification.
 - Developed a custom **Trainer with weighted loss functions**, improving minority class **performance by 25%**.
 - Future work on **Hindi LLMs** using both encoder-only and encoder-decoder setups for poem generation.**Keywords:** *Hugging Face, IndicBERT, LLM Fine-tuning, Text Generation, NLP, Poetry Generation*
- **DoodleVision** — Prof. Priyanka Bagade, IIT Kanpur (Feb 2025 – Present)
 - Built novel hybrid model fusing image+stroke data; achieved **89.41% accuracy**, **+5%** over benchmark .
 - Paper under preparation for **WACV 2026**, highlighting novelty in architecture and efficient multimodal fusion.
 - Designed disk-streaming pipeline for **memory-constrained environments**, training with 25M+ examples.
 - Deploying optimized inference pipeline to edge devices (Raspberry Pi) for real-time applications.**Keywords:** *Deep Learning, Multimodal Fusion, Disk-Streaming, Edge Deployment, CNN, LSTM*
- **EarlAz: Early Detection of Alzheimer's using Mobile Game and ML** — Prof. Veena Bharti, ABESIT (Jan 2022 – Jun 2022)
 - Designed a mobile game to collect cognitive-behavioral data for early-stage Alzheimer's detection.
 - Implemented SVM and Decision Tree classifiers to predict cognitive decline from gameplay features.**Keywords:** *Unity3D, SVM, Decision Tree, Cognitive Games, Keras, GPU*

PROJECTS

- **WhatsApp Key/Database Extractor**  — (Self Project, Open Source) (Nov 2020 – Aug 2023)
 - Engineered a **forensic tool** to **remotely** extract WhatsApp keys & databases from Android devices.
 - Enabled **migration of WhatsApp chats to Signal**, collaborating with Swiss developer on  Signal-Android.
 - Open-sourced with strong adoption: **520+ stars**, **70+ forks**, **500K+ hits** on GitHub.**Keywords:** *Python, Android, Digital Forensics, Security Research, Android Debug Bridge, Collaboration*

- Blockchain-based Ticketing System**  — CS731 Blockchain & it's Applications (Mar 2025 – Apr 2025)
Mentor: Prof. Angshuman Karmakar, IIT Kanpur
 - Developed a blockchain ticketing system with web UI for secure booking, cancellation, and rescheduling.
 - Implemented smart contracts for payments, verifiable records, and double-booking & overbooking prevention.**Keywords:** Hyperledger Fabric, Smart Contracts, Blockchain, Node.js, Kubernetes, Docker
- Detection of Deepfake**  — CS685 Data Mining (Sep 2024 – Nov 2024)
Mentor: Prof. Arnab Bhattacharya, IIT Kanpur
 - Built a web platform to let users review videos as authentic/deepfake; aggregated responses for analysis.
 - Studied patterns, biases, and temporal trends in human judgment of deepfakes.**Keywords:** Deepfake Detection, User Analytics, Thematic Analysis, TF-IDF, Temporal Analysis
- Unsupervised Domain Adaptation**  — CS771 - Introduction to Machine Learning (Aug 2024 – Nov 2024)
Mentor: Prof. Piyush Rai, IIT Kanpur
 - Built a pipeline to evaluate **SVM adaptability** on evolving unlabeled streams using pseudo-labeling and re-clustering.
 - Automated ingestion, incremental learning, and visual diagnostics for monitoring across **shifting domains**.**Keywords:** Python, NumPy, scikit-learn, Pandas, Matplotlib, Domain Adaptation
- AmazoLens**  — CS661 Big Data Visual Analytics (Feb 2025 – Apr 2025)
Mentor: Prof. Soumya Dutta, IIT Kanpur
 - Built a dashboard for Amazon data to analyze sales, segment customers, and recommend products with ML.
 - Applied market basket analysis, topic mining, sentiment analysis, and forecasting (Prophet).**Keywords:** Data Visualization, Market Basket Analysis, NLP, Sentiment Analysis, Recommendation, D3.js

TECHNICAL SKILLS

- Languages:** Python, C/C++, JavaScript, SQL, C#
 DevOps/Cloud: Git, Docker, Kubernetes, AWS, Sage-Maker, GCP, Azure, Terraform, CI/CD
- Web:** FastAPI, Flask, Express.js, Node.js, React, GraphQL, gRPC, REST
 Others: Kafka, Zookeeper, RabbitMQ, Airflow, Spark, Hadoop, Hyperledger Fabric, Bash, YAML, L^AT_EX
- Databases:** PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch
- ML/Frameworks:** NumPy, Pandas, TensorFlow, PyTorch, Scikit-learn, Hugging Face, OpenCV

POSITIONS OF RESPONSIBILITY

- Teaching Assistant for CS787 Generative Artificial Intelligence, CSE, IIT Kanpur. (Jul 2025 – Present)
- Teaching Assistant for ESC111/2, CSE Department, IIT Kanpur. (Jan 2025 – May 2025)
- Teaching Assistant for ELC courses, CSE Department, IIT Kanpur. (July 2024 –Nov 2024)

SCHOLASTIC ACHIEVEMENTS

- Ranked **157th/123,967** in *GATE 24 (CS)*.
- Ranked **724th/39,210** in *GATE 24 (DS & AI)*.
- Earned **200 / 100 / 50-day streaks** on Leetcode, plus **Monthly Medals** (Oct 2024 – Jan 2025).
- Graduated with **1st Division with Honours**, *ABES Institute of Technology*.

RELEVANT COURSES

Data Structures & Algorithms	Operating Systems	Computer Networks*
Database Management Systems	Compiler Design	Software Engineering
Discrete Mathematics	Theory of Computation	Cryptography & Network Security
Intro to Machine Learning (CS771)	Data Mining (CS685)*	Deep Learning for CV (CS776)
Big Data Visual Analytics (CS661)	Cloud Computing	Blockchain & Applications (CS731)*
Computational Geometry (CS663)	Data Analytics	Discrete Structures & Logic

* Secured grade A